Technical Supplement: Limiting Consistency of Smooth Penalty Prices with Hard-Constrained KKT Multipliers

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Abstract

This supplement gives the detailed argument connecting the smooth voltage and congestion scarcity prices used in the accompanying manuscript to the KKT multipliers of its centralized hard-constrained LinDistFlow OPF. The result is an iterated limiting statement along the minimizers of the corresponding smooth and quadratic-penalty problems. It is not a pointwise identification at an arbitrary submitted power vector, and finite-parameter scarcity prices are not exact hard-OPF dual variables. Fixed load coordinates are eliminated before the constraint qualification is imposed, which avoids the degeneracy of representing a zero-width interval by two active bound inequalities.

1 Effective-space hard-constrained problem

1.1 Elimination of rigid coordinates

Let the flexible coordinates be collected in $\mathbf{y} \in \mathcal{X} \subset \mathbb{R}^d$, where

$$\mathcal{X} := \prod_{h=1}^{d} [\underline{y}_h, \overline{y}_h], \qquad \underline{y}_h < \overline{y}_h. \tag{1}$$

Rigid loads are embedded in the full non-root-bus power vector through

$$\mathbf{p}(\mathbf{y}) = \mathbf{p}^{\text{fix}} + G_a \mathbf{y}. \tag{2}$$

Thus, no zero-width interval is retained as a pair of active inequalities in the optimization coordinates. This point matters for the linear-independence constraint qualification used below.

Define the reduced centralized objective

$$F(\mathbf{y}) := \mathcal{S}(\mathbf{p}(\mathbf{y})), \tag{3}$$

where $\mathcal{S}$ is the welfare objective in the manuscript. Under its main-study assumptions,

$$\nabla^2 F(\mathbf{y}) = D_{\text{util}} + \frac{2\pi^E}{v_0} G_a^\mathsf{T} H^L G_a \succ 0, \tag{4}$$

because $D_{\text{util}} = \operatorname{diag}(\beta_{u,h}) \succ 0$, $\pi^E \geq 0$, and $H^L \succeq 0$. Hence F is continuously differentiable and strongly convex on $\mathcal{X}$.

For the main-study lower-voltage and forward-flow limits, set

$$g_m^V(\mathbf{y}) := \underline{v} - v_m(\mathbf{p}(\mathbf{y})), \tag{5}$$

$$g_\ell^C(\mathbf{y}) := f_\ell(\mathbf{p}(\mathbf{y})) - \overline{f}_\ell. \tag{6}$$

Stack all network inequalities as $g_i(\mathbf{y}) \leq 0$, $i = 1, \ldots, r$. These functions are affine. Let

$$G := [\nabla g_1 \ \cdots \ \nabla g_r] \in \mathbb{R}^{d \times r}. \tag{7}$$

The hard-constrained effective-space problem is

$$\min_{\mathbf{y} \in \mathcal{X}} F(\mathbf{y}) \quad \text{subject to} \quad g(\mathbf{y}) \leq 0. \tag{8}$$

1.2 KKT multipliers and scarcity-price mapping

Assume that problem (8) is feasible, and denote its unique minimizer by $\mathbf{y}^\star$. The uniqueness follows from Eq. (4). Let

$$\mathcal{A}_N^\star := \{i : g_i(\mathbf{y}^\star) = 0\} \tag{9}$$

be the active network-constraint set. Let $B^\star$ collect one outward normal for every active bound of the nondegenerate box $\mathcal{X}$ at $\mathbf{y}^\star$. We impose the effective-space LICQ

$$[\, B^\star \quad G_{\mathcal{A}_N^\star} \,] \quad \text{has full column rank,} \tag{10}$$

where $G_{\mathcal{A}_N^\star}$ contains the columns of G indexed by $\mathcal{A}_N^\star$. Unlike a full-space LICQ imposed on two inequalities for a zero-width interval, Eq. (10) is compatible with rigid loads because those coordinates have already been eliminated.

The hard KKT system is

$$0 \in \nabla F(\mathbf{y}^\star) + G\boldsymbol{\omega}^{\mathrm{h}} + N_{\mathcal{X}}(\mathbf{y}^\star), \tag{11}$$

$$g(\mathbf{y}^\star) \le 0, \qquad \boldsymbol{\omega}^{\mathrm{h}} \ge 0, \tag{12}$$

$$\omega_i^{\mathrm{h}} g_i(\mathbf{y}^\star) = 0, \qquad i = 1, \dots, r. \tag{13}$$

Here $N_{\mathcal{X}}$ is the convex-analysis normal cone. Condition (10) makes the complete KKT multiplier vector, and hence the network multiplier vector $\boldsymbol{\omega}^{\mathrm{h}}$, unique.

Partition $\boldsymbol{\omega}^{\mathrm{h}}$ into the low-voltage multipliers $\boldsymbol{\lambda}^{\mathrm{h}}$ and the forward-flow multipliers $\boldsymbol{\nu}^{\mathrm{h}}$. Since

$$\frac{\partial g_m^V}{\partial p_k} = A_{mk}^V, \qquad \frac{\partial g_\ell^C}{\partial p_k} = A_{\ell k}^C, \tag{14}$$

the corresponding full-power marginal terms are

$$\pi_k^{V,\star} = \sum_m A_{mk}^V \lambda_m^{\mathrm{h}}, \qquad \pi_k^{C,\star} = \sum_\ell A_{\ell k}^C \nu_\ell^{\mathrm{h}}. \tag{15}$$

This establishes the sign and sensitivity structure used in the hard DLMP.

2 Quadratic and smooth outer-penalty problems

2.1 Positive-part smoothing

For $z \in \mathbb{R}$, let

$$q(z) := \frac{1}{2}[z]_+^2, \qquad s_\varepsilon(z) := \varepsilon \log(1 + e^{z/\varepsilon}), \qquad \varphi_\varepsilon(z) := \int_{-\infty}^{z} s_\varepsilon(t)\, dt. \tag{16}$$

The function q is continuously differentiable with $q'(z) = [z]_+$, whereas $\varphi_\varepsilon \in C^\infty$ and $\varphi_\varepsilon' = s_\varepsilon$. The standard Softplus bounds are

$$0 \le s_\varepsilon(z) - [z]_+ \le \varepsilon \log 2, \qquad z \in \mathbb{R}. \tag{17}$$

Moreover,

$$0 \le \varphi_\varepsilon(z) - q(z) \le \frac{\pi^2}{6}\varepsilon^2, \qquad z \in \mathbb{R}. \tag{18}$$

Indeed, the derivative of the difference in Eq. (18) is nonnegative, and its total increase over $\mathbb{R}$ is

$$2\varepsilon^2 \int_0^\infty \log(1 + e^{-u})\, du = \frac{\pi^2}{6}\varepsilon^2. \tag{19}$$

These bounds also show why smoothing errors in the multiplier coefficients are scaled by the penalty parameter.

2.2 Penalized minimizers and multiplier estimates

Let $\rho_i > 0$ be fixed weights. In the manuscript, $\kappa_V = \rho_V \kappa$ and $\kappa_C = \rho_C \kappa$, so each ρ_i is either ρ_V or ρ_C. The quadratic outer-penalty problem is

$$\mathbf{y}^{(\kappa)} := \operatorname*{argmin}_{\mathbf{y}\in\mathcal{X}} \left\{ F(\mathbf{y}) + \kappa \sum_{i=1}^{r} \rho_i q(g_i(\mathbf{y})) \right\}. \tag{20}$$

Its multiplier estimates are

$$\omega_i^{(\kappa)} := \kappa\rho_i [g_i(\mathbf{y}^{(\kappa)})]_+. \tag{21}$$

The smooth outer-penalty problem is

$$\mathbf{y}^{(\kappa,\varepsilon)} := \operatorname*{argmin}_{\mathbf{y}\in\mathcal{X}} \left\{ F(\mathbf{y}) + \kappa \sum_{i=1}^{r} \rho_i \varphi_\varepsilon(g_i(\mathbf{y})) \right\}, \tag{22}$$

with smooth multiplier estimates

$$\omega_i^{(\kappa,\varepsilon)} := \kappa\rho_i s_\varepsilon(g_i(\mathbf{y}^{(\kappa,\varepsilon)})). \tag{23}$$

Strong convexity of F, convexity of q and φ_ε, and affinity of g_i make both penalized minimizers unique. Their stationarity systems are

$$0 \in \nabla F(\mathbf{y}^{(\kappa)}) + G\boldsymbol{\omega}^{(\kappa)} + N_{\mathcal{X}}(\mathbf{y}^{(\kappa)}), \tag{24}$$
$$0 \in \nabla F(\mathbf{y}^{(\kappa,\varepsilon)}) + G\boldsymbol{\omega}^{(\kappa,\varepsilon)} + N_{\mathcal{X}}(\mathbf{y}^{(\kappa,\varepsilon)}). \tag{25}$$

For voltage and congestion constraints, Eq. (23) is exactly

$$\mu_m^{(\kappa,\varepsilon)} = \kappa_V s_\varepsilon\Big(\underline{v} - v_m(\mathbf{p}(\mathbf{y}^{(\kappa,\varepsilon)}))\Big), \qquad \nu_\ell^{(\kappa,\varepsilon)} = \kappa_C s_\varepsilon\Big(f_\ell(\mathbf{p}(\mathbf{y}^{(\kappa,\varepsilon)})) - \overline{f}_\ell\Big). \tag{26}$$

3 Convergence results

3.1 Removal of the smoothing error

Proposition 1 (Fixed-κ smoothing limit)**.** *For every fixed $\kappa > 0$,*

$$\mathbf{y}^{(\kappa,\varepsilon)} \longrightarrow \mathbf{y}^{(\kappa)}, \qquad \boldsymbol{\omega}^{(\kappa,\varepsilon)} \longrightarrow \boldsymbol{\omega}^{(\kappa)} \quad \text{as } \varepsilon \downarrow 0. \tag{27}$$

Proof. By Eq. (18), the two objectives in Eqs. (20) and (22) satisfy the uniform estimate

$$0 \le P_{\kappa,\varepsilon}(\mathbf{y}) - P_\kappa(\mathbf{y}) \le \frac{\pi^2}{6}\kappa\varepsilon^2 \sum_{i=1}^{r} \rho_i, \qquad \mathbf{y} \in \mathcal{X}. \tag{28}$$

Thus $P_{\kappa,\varepsilon} \to P_\kappa$ uniformly on the compact set $\mathcal{X}$. Every accumulation point of $\mathbf{y}^{(\kappa,\varepsilon)}$ minimizes P_κ. Since this minimizer is unique, the entire sequence converges to $\mathbf{y}^{(\kappa)}$.

For each i, Eq. (17) and the 1-Lipschitz property of $[\cdot]_+$ give

$$\begin{aligned}\left|\omega_i^{(\kappa,\varepsilon)} - \omega_i^{(\kappa)}\right| &\le \kappa\rho_i\varepsilon \log 2 \\ &\quad + \kappa\rho_i \left| g_i(\mathbf{y}^{(\kappa,\varepsilon)}) - g_i(\mathbf{y}^{(\kappa)})\right|. \end{aligned} \tag{29}$$

Affinity of g_i and convergence of the minimizers make the right-hand side tend to zero. □

3.2 Quadratic-penalty limit

Proposition 2 (Quadratic outer-penalty consistency)**.** *Under the assumptions in Section 1, including the effective-space LICQ in Eq.* (10)*,*

$$\mathbf{y}^{(\kappa)} \longrightarrow \mathbf{y}^{\star}, \qquad \boldsymbol{\omega}^{(\kappa)} \longrightarrow \boldsymbol{\omega}^{\mathrm{h}} \quad \textit{as } \kappa \to \infty. \tag{30}$$

Proof. Because $\mathbf{y}^{\star}$ is feasible for the hard problem, optimality of $\mathbf{y}^{(\kappa)}$ yields

$$F(\mathbf{y}^{(\kappa)}) + \frac{\kappa}{2}\sum_{i=1}^{r} \rho_i [g_i(\mathbf{y}^{(\kappa)})]_+^2 \le F(\mathbf{y}^{\star}). \tag{31}$$

Let $F_{\min} := \min_{\mathbf{y}\in\mathcal{X}} F(\mathbf{y})$ and $\rho_{\min} := \min_i \rho_i > 0$. Compactness of $\mathcal{X}$ and continuity of F ensure that $F_{\min}$ is finite. Equation (31) implies

$$\sum_{i=1}^{r} [g_i(\mathbf{y}^{(\kappa)})]_+^2 \le \frac{2\big(F(\mathbf{y}^{\star}) - F_{\min}\big)}{\kappa \rho_{\min}} \longrightarrow 0. \tag{32}$$

Every accumulation point of $\mathbf{y}^{(\kappa)}$ is therefore hard-feasible. Also, Eq. (31) gives $F(\mathbf{y}^{(\kappa)}) \le F(\mathbf{y}^{\star})$. Hence every accumulation point is a global minimizer of the hard problem. Its uniqueness implies $\mathbf{y}^{(\kappa)} \to \mathbf{y}^{\star}$.

It remains to prove multiplier convergence. If $i \notin \mathcal{A}_N^{\star}$, then $g_i(\mathbf{y}^{\star}) < 0$. Convergence of the primal sequence implies $g_i(\mathbf{y}^{(\kappa)}) < 0$ and consequently $\omega_i^{(\kappa)} = 0$ for all sufficiently large κ.

Let Z have orthonormal columns spanning the tangent subspace of the active box face at $\mathbf{y}^{\star}$:

$$\mathrm{range}(Z) = \ker((B^{\star})^{\mathsf{T}}). \tag{33}$$

Any bound inactive at $\mathbf{y}^{\star}$ remains inactive at $\mathbf{y}^{(\kappa)}$ for sufficiently large κ. On a coordinate active at $\mathbf{y}^{\star}$, the penalty minimizer is either interior or on the same bound once it is sufficiently close to $\mathbf{y}^{\star}$. Therefore, for every sufficiently large κ and every $\mathbf{n}^{(\kappa)} \in N_{\mathcal{X}}(\mathbf{y}^{(\kappa)})$ appearing in Eq. (24),

$$\mathbf{n}^{(\kappa)} \in \mathrm{range}(B^{\star}), \qquad Z^{\mathsf{T}}\mathbf{n}^{(\kappa)} = 0. \tag{34}$$

Set

$$M := Z^{\mathsf{T}} G_{\mathcal{A}_N^{\star}}. \tag{35}$$

The effective-space LICQ makes M full column rank. To see this, if $M\mathbf{c} = 0$, then $G_{\mathcal{A}_N^{\star}}\mathbf{c} \in \mathrm{range}(B^{\star})$; full column rank of $[B^{\star}\ G_{\mathcal{A}_N^{\star}}]$ then forces $\mathbf{c} = 0$.

Projecting Eq. (24) by Z^{T} and using Eqs. (34)–(35) gives

$$Z^{\mathsf{T}}\nabla F(\mathbf{y}^{(\kappa)}) + M\boldsymbol{\omega}_{\mathcal{A}_N^{\star}}^{(\kappa)} = 0. \tag{36}$$

Consequently,

$$\boldsymbol{\omega}_{\mathcal{A}_N^{\star}}^{(\kappa)} = -(M^{\mathsf{T}}M)^{-1}M^{\mathsf{T}}Z^{\mathsf{T}}\nabla F(\mathbf{y}^{(\kappa)}). \tag{37}$$

Taking $\kappa \to \infty$ and using $\mathbf{y}^{(\kappa)} \to \mathbf{y}^{\star}$ gives the unique solution of the projected hard-KKT stationarity equation. By Eq. (10), that solution is exactly $\boldsymbol{\omega}_{\mathcal{A}_N^{\star}}^{\mathrm{h}}$. Together with the eventual zeros outside $\mathcal{A}_N^{\star}$, this proves convergence of the full network multiplier vector. □

3.3 Iterated smooth-to-hard limit

Theorem 1 (Limiting consistency of the smooth scarcity prices)**.** *Under the assumptions in Section 1,*

$$\lim_{\kappa\to\infty}\left(\lim_{\varepsilon\downarrow 0}\mathbf{y}^{(\kappa,\varepsilon)}\right)=\mathbf{y}^{\star}, \tag{38}$$

and

$$\lim_{\kappa\to\infty}\left(\lim_{\varepsilon\downarrow 0}\boldsymbol{\omega}^{(\kappa,\varepsilon)}\right)=\boldsymbol{\omega}^{\mathrm{h}}. \tag{39}$$

Accordingly,

$$\lim_{\kappa\to\infty}\lim_{\varepsilon\downarrow 0}(A^{V})^{\mathsf{T}}\boldsymbol{\mu}^{(\kappa,\varepsilon)}=(A^{V})^{\mathsf{T}}\boldsymbol{\lambda}^{\mathrm{h}}, \tag{40}$$

$$\lim_{\kappa\to\infty}\lim_{\varepsilon\downarrow 0}(A^{C})^{\mathsf{T}}\boldsymbol{\nu}^{(\kappa,\varepsilon)}=(A^{C})^{\mathsf{T}}\boldsymbol{\nu}^{\mathrm{h}}. \tag{41}$$

Proof. Apply Proposition 1 for each fixed κ, and then apply Proposition 2. The price-component limits follow because A^V and A^C are fixed linear maps. □

The two propositions are model-specific versions of standard smoothing and quadratic exterior-penalty results; see Refs. [1, 2, 3] and the variational treatment in Ref. [4].

4 Scope of the limiting statement

The evaluation points in Theorem 1 are essential. The result does not assert that

$$\kappa\rho_i[g_i(\mathbf{y})]_+\longrightarrow\omega_i^{\mathrm{h}} \tag{42}$$

for a fixed arbitrary $\mathbf{y}$. In particular, at a binding hard solution $g_i(\mathbf{y}^\star)=0$, the left-hand side of Eq. (42) is zero, even when $\omega_i^{\mathrm{h}}>0$. The nonzero multiplier is recovered from the $O(1/\kappa)$ constraint displacement of the quadratic-penalty minimizer.

For example, consider

$$\min_{y\in[-1,1]}\frac{1}{2}(y-1)^2\quad\text{subject to}\quad y\le 0. \tag{43}$$

The hard solution is $y^\star=0$ with multiplier $\omega^{\mathrm{h}}=1$. The quadratic-penalty solution is

$$y^{(\kappa)}=\frac{1}{1+\kappa},\qquad \omega^{(\kappa)}=\kappa y^{(\kappa)}=\frac{\kappa}{1+\kappa}\longrightarrow 1. \tag{44}$$

Direct evaluation at $y^\star$, in contrast, gives $\kappa[y^\star]_+=0$.

The order of the two regularizations also matters. From Eq. (17), at a common argument z,

$$0\le\kappa\rho_i s_\varepsilon(z)-\kappa\rho_i[z]_+\le\kappa\rho_i\varepsilon\log 2. \tag{45}$$

Thus $\varepsilon\downarrow 0$ at fixed κ removes the smoothing error, but an arbitrary joint limit of (κ,ε) is not implied by Theorem 1. A joint-limit statement would require an additional scaling condition and a separate proof.

Finally, Theorem 1 concerns the minimizers of the centralized hard and penalized OPFs. At an arbitrary submitted power vector, or at a QNE of the responsibility-aware game, the finite-parameter coefficients remain explicit smooth scarcity signals. They should not be described as exact hard-OPF KKT multipliers. Their finite-parameter agreement with the hard centralized benchmark is therefore an empirical accuracy question, as assessed in the manuscript's numerical price-verification study.

References

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